



CME 3204 Data Communications and Computer Networks

Midterm Project Report

Metropolitan Area Network Simulation

Using Cisco Packet Tracer

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Abstract

This project report details the topological design, architecture, and traffic characterization of a Metropolitan Area Network (MAN) using Cisco Packet Tracer. The primary objective is to build a robust and scalable network connecting two distinct branch offices in a city via an Internet Service Provider (ISP) infrastructure. The design accommodates diverse facility requirements, integrating end-devices, wireless networks, VoIP capabilities, and a dedicated server farm to ensure maximum availability and minimum delay. Through discrete-event simulation, nine distinct network traffic scenarios are analyzed to validate network functionality, including routing protocols, access controls, and application-layer services (HTTP, FTP, DNS, DHCP, and SSH). The results demonstrate a successful balance between system requirements and network architecture.

Table of Contents

Contents

1. Introduction.....	5
1.1. Project Definition and Problem Formulation	5
1.2. Purpose and Motivation.....	5
1.3. Related Work	5
2. Method and Simulation	6
2.1. Simulation and Modeling Concepts	6
2.2. Simulation Environment – Cisco Packet Tracer.....	6
2.3. Network Design Requirements.....	6
2.4. Requirement Analysis	7
2.5. System/Model Definitions	7
2.6. Simulation Elements	7
3. Traffic Analysis and Simulation Results	8
3.1. Scenario 1: Wireless Web & Email Access	8
3.2. Scenario 2: FTP File Transfer.....	10
3.3. Scenario 3: VoIP Conference Call	12
3.4. Scenario 4: Inter-Branch Email	13
3.5. Scenario 5: ICMP Ping – Cross-Branch	16
3.6. Scenario 6: Wireless Laptop Email	18
3.7. Scenario 7: SSH Remote Management	20
3.8. Scenario 8: DNS Domain Resolution & Web Service Access	21
3.9. Scenario 9: Round-Robin DNS Load Balancing – FTP Servers	24
4. Conclusion.....	24
5. References	25

Figure Table

Figure 1: MAN Simulation Topology in Cisco Packet Tracer	6
Figure 3.1.1: Laptop0 obtaining IP addresses from DHCP	8
Figure 3.1.2: Laptop1 obtaining IP addresses from DHCP	8
Figure 3.1.3: Laptop2 obtaining IP addresses from DHCP	8
Figure 3.1.4: DNS Server Records for Branch 1 Facility 3	9
Figure 3.2.1: FTP packet initiation from Branch 2, Facility 2	10
Figure 3.2.2: FTP packet arrival at Branch 1, Facility 3	10
Figure 3.2.3: Simulation Event List for FTP file transfer	11
Figure 3.3.1: VoIP call establishment between IP Phone 0 and IP Phone 1	10
Figure 3.4.1: Centralized Mail Server Domain Configuration	11
Figure 3.4.2: Email Compose Interface on PC2	12
Figure 3.4.3: DNS Resolution and SMTP Send Success Status	12
Figure 3.4.4: Mail Browser Interface on Recipient PC11	13
Figure 3.4.5: POP3 Retrieval and Receive Mail Success Status	13
Figure 3.5.1: Target Web Server IP Configuration (192.168.30.10)	14
Figure 3.5.2: Ping command execution in Command Prompt	14
Figure 3.5.3: ICMP packet generation at the source device	15
Figure 3.5.4: ICMP packet arrival at the destination web server	15
Figure 3.6.1: Email Client Configuration on Laptop3	16
Figure 3.6.2: Successful DNS Resolution and SMTP session for wireless client	17
Figure 3.6.3: Successful Email Retrieval on Smartphone2 via POP3	17
Figure 3.7.1: SSH Client Configuration and Host Selection	18
Figure 3.7.2: Secure SSH Session established on MerkezRouter	18
Figure 3.8.1: Web Browser request to http://merkezi.com	19
Figure 3.8.2: DNS Query packet initiation at PC23	19
Figure 3.8.3: DNS Server returning IP address from Server Farm	20
Figure 3.8.4: HTTP packet return to the source device	20
Figure 3.8.5: Web service access flow in Simulation Event List	21
Figure 3.9.1: Nslookup command results for ftp.merkezi.com	22
Figure 3.9.2: Initial DNS request for FTP load balancing	22
Figure 3.9.3: Traffic distribution to available FTP servers	23
Figure 3.9.4: Inbound PDU Details showing multiple DNS Answer records	23

1. Introduction

Network planning and design is a highly iterative process that requires balancing the cost of network hardware with systemic functional requirements. This chapter introduces the problem formulation, core motivations, and the theoretical background necessary for deploying a simulated Metropolitan Area Network.

1.1. Project Definition and Problem Formulation

The core problem addressed in this project is the design and simulation of a Metropolitan Area Network (MAN). The network must connect two distinct branch offices located within the same city, interconnected by at least two routers per branch over an Internet Service Provider (ISP) network. The formulation requires careful IP subnetting, routing protocol configuration, and the deployment of distinct sub-networks across six separate facilities (three per branch), each presenting unique device densities and functional requirements.

1.2. Purpose and Motivation

The motivation behind this project is to model a real-world enterprise network expansion. In modern corporate environments, distinct branches require seamless, secure, and fast communication to share centralized resources (such as server farms). The primary business goal is to establish an architecture that supports high traffic loads, provides real-time communication (VoIP), and offers adequate hardware support for future expansions. By realizing this project, the enterprise benefits from centralized data management and ubiquitous connectivity, while mitigating the risks of broadcast storms and unauthorized access through proper logical segmentation (VLANs) and routing.

1.3. Related Work

Network modeling and simulation are critical precursors to physical network deployment. Previous studies highlight that simulating network traffic before physical implementation prevents costly hardware bottlenecks. The principles of discrete-event simulation have been widely adopted to evaluate queueing delays, packet loss, and routing efficiencies in MAN architectures.

2. Method and Simulation

This chapter outlines the network requirements, constraints, and the simulation methodologies considered during the modeling phase.

2.1. Simulation and Modeling Concepts

This project relies on discrete-event simulation. In a discrete-event system, state changes occur at distinct points in time triggered by events, such as a packet arriving at a router interface or a TCP handshake initiation. The primary benefit of utilizing modeling and simulation over physical implementations is the ability to test hypothetical traffic loads, configure complex routing algorithms safely, and observe packet-level behaviors without the financial and operational risks associated with live network manipulation.

2.2. Simulation Environment – Cisco Packet Tracer

- **Architecture and Modeling Concepts:** Packet Tracer uses a split-architecture approach, offering both a logical workspace for topology design and a physical workspace for hardware layout.
- **Capabilities and Limitations:** It perfectly simulates OSI Layers 1 through 7, providing detailed PDU inspections. However, it is limited in simulating advanced proprietary ISP core routing protocols or deep packet inspection (DPI) firewalls.
- **Modules and Components:** The project uses 2911 Series Routers, 2960 Series Switches, generic Servers (HTTP, DNS, FTP, DHCP, Mail), Wireless Access Points, and various end-user devices (PCs, Laptops, Smartphones, Tablets).

2.3. Network Design Requirements

The network is structured hierarchically. The backbone consists of an ISP connecting the edge routers of the two branches.

Network Topology – Cisco Packet Tracer

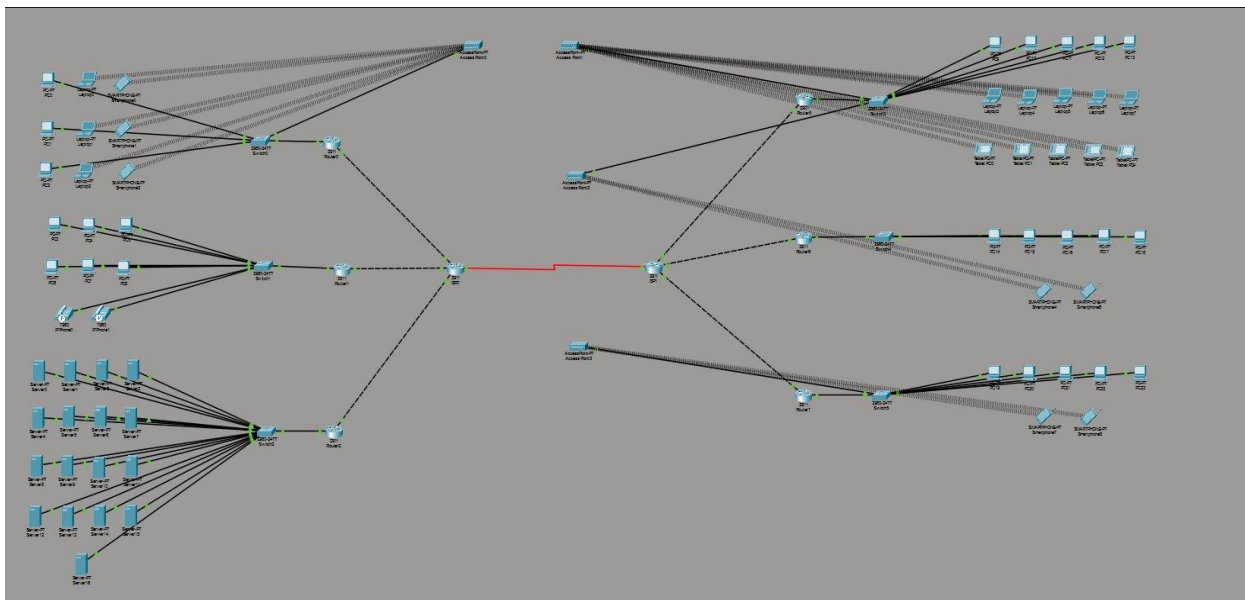


Figure 1: MAN Simulation Topology in Cisco Packet Tracer

Branch 1 – Facility Overview

Facility	Devices	Services
Facility 1	3 PCs, 3 Laptops, 3 Smartphones	Web, Email, FTP
Facility 2	6 PCs (2 dedicated VoIP)	Web, FTP, VoIP Conference
Facility 3 (Server Farm)	10 Web, 4 FTP, 1 DHCP, 1 Mail, 1 DNS Server	All application services

Branch 2 – Facility Overview

Facility	Devices	Services
Facility 1	5 PCs, 5 Laptops, 5 Tablets (Wireless)	Web, Email, Internet
Facility 2	5 PCs, 2 Smartphones	Web, App Editing, FTP
Facility 3	5 PCs, 2 Mobile Devices	Web, Email

2.4. Requirement Analysis

- Functional requirements: The network must successfully route VoIP (RTP/SIP) for Branch 1 Facility 2, support robust FTP, and provide unified DNS resolution across the entire MAN.
- Performance requirements: The network must support 56 diverse endpoints simultaneously without severe latency, utilizing FastEthernet/GigabitEthernet links.
- Constraints: Broadcast domains must be minimized. Default routing and NAT over the ISP link must be efficiently configured.

2.5. System/Model Definitions

- Assumptions: Intra-facility traffic uses Star topology; inter-facility traffic uses a Tree/Hierarchical topology.
- Addressing and Routing: IPv4 scheme — 192.168.X.0/24 for local subnets, 10.0.0.X/30 for WAN links. Dynamic routing (OSPF or EIGRP) or static inter-branch routing is used.
- Device Types: Managed Cisco devices form the core; heterogeneous end-nodes (PCs, tablets, smartphones) are the data sources.

2.6. Simulation Elements

System Entities	Computer nodes, server farm, router interfaces, queues, packet flows
State Variables	Interface status (up/down/idle/busy), routing table convergence state
Input Variables	Packet arrival rate, user commands (ping, ftp)
Resources	Link bandwidth, DHCP IP pool, number of active servers
Activities & Events	ICMP echo, queuing delays, TCP three-way handshakes, packet arrivals

3. Traffic Analysis and Simulation Results

This section demonstrates the network functionality and provides PDU analysis for the mandatory and custom simulation scenarios. Screenshots from Cisco Packet Tracer's Simulation Mode and PDU Information Window should be inserted in the designated placeholders below.

Scenario 1: Wireless Web & Email Access

Branch 2, Facility 1 – Laptop wirelessly browses the web and reads emails.

Network Functionality:

The laptop successfully acquires an IP via DHCP, resolves the Web/Mail server IP via DNS, and establishes HTTP/SMTP/POP3 connections through the access point and branch router to the server farm in Branch 1, Facility 3.

Protocol Data Units (PDU) Content:

Figure 3.1.1 Laptop0 obtaining IP addresses from DHCP.

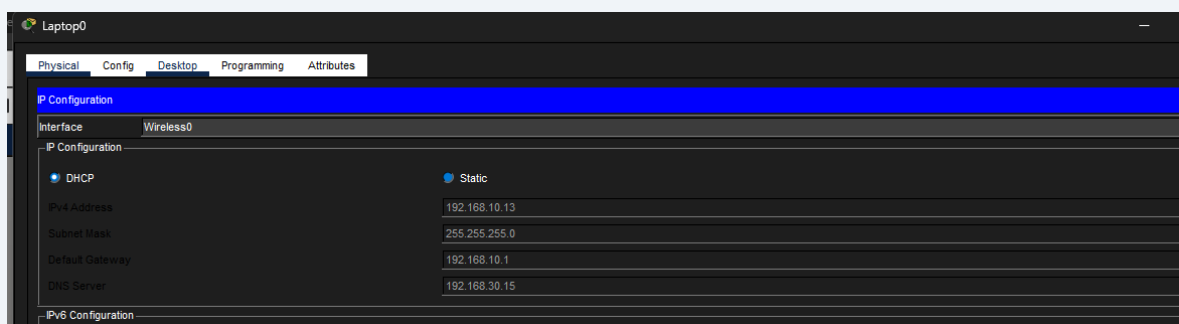


Figure 3.1.2 Laptop1 obtaining IP addresses from DHCP.

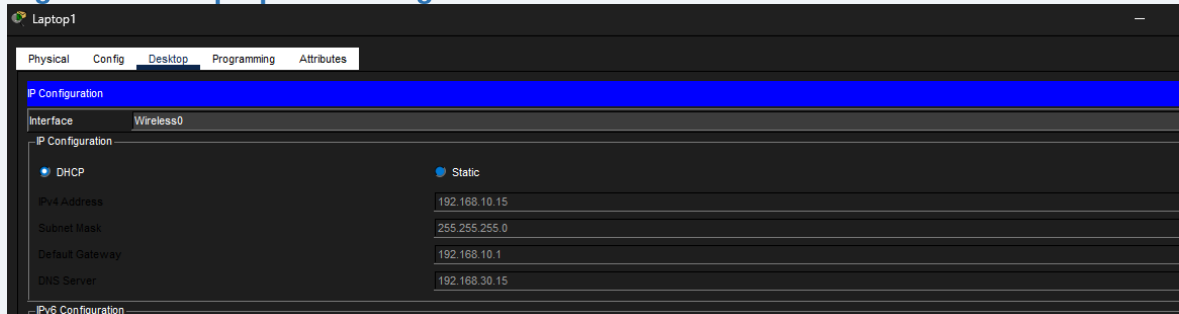


Figure 3.1.3 Laptop2 obtaining IP addresses from DHCP.

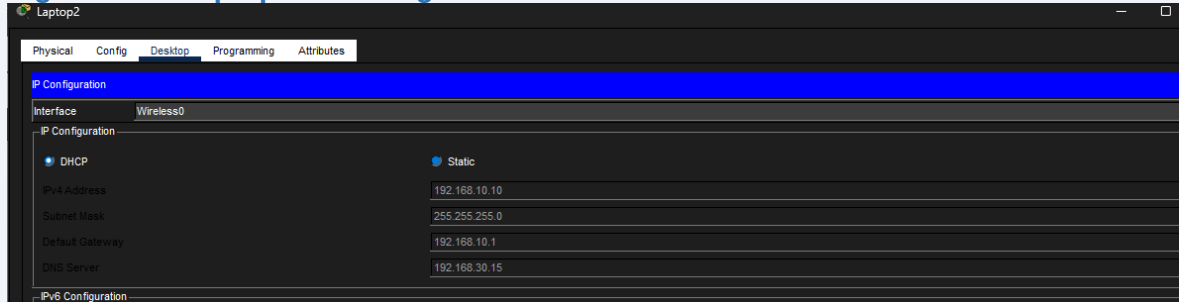
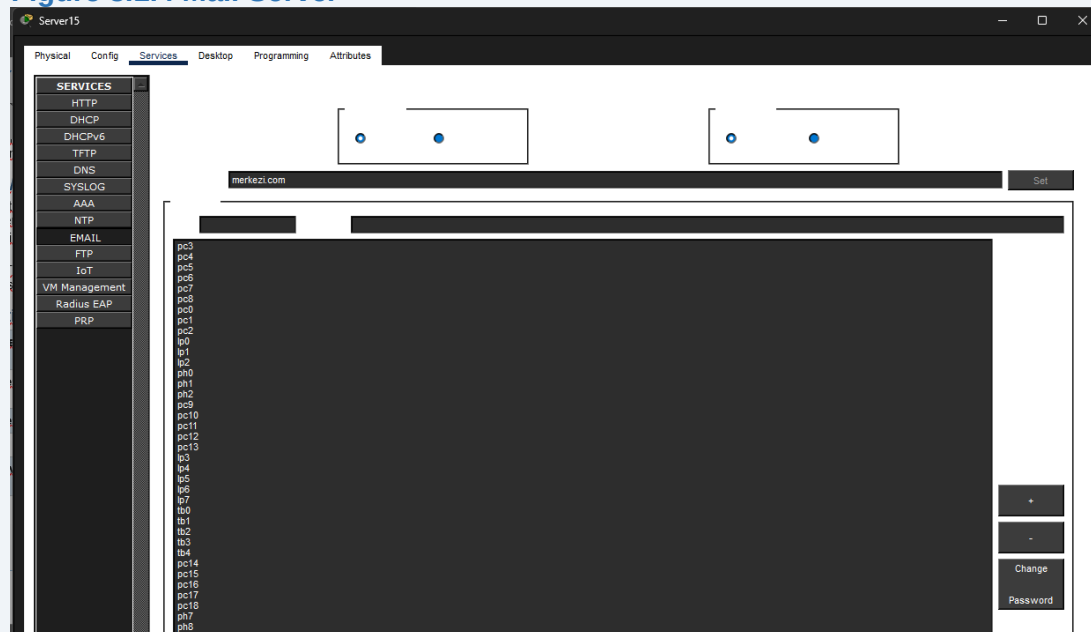


Figure 3.1.3 After the laptops in branch 1, facility 3 receive IP addresses via DHCP, their mail addresses connect to the mail server via the DNS server.

No.	Name	Type	Detail
0	ftp.merkezi.com	A Record	192.168.30.21
1	ftp.merkezi.com	A Record	192.168.30.22
2	ftp.merkezi.com	A Record	192.168.30.23
3	ftp.merkezi.com	A Record	192.168.30.24
4	mail.merkezi.com	A Record	192.168.30.25
5	merkezi.com	A Record	192.168.30.10
6	merkezi.com	A Record	192.168.30.11
7	merkezi.com	A Record	192.168.30.12
8	merkezi.com	A Record	192.168.30.13
9	merkezi.com	A Record	192.168.30.14
10	merkezi.com	A Record	192.168.30.15
11	merkezi.com	A Record	192.168.30.16
12	merkezi.com	A Record	192.168.30.17
13	merkezi.com	A Record	192.168.30.18
14	merkezi.com	A Record	192.168.30.19
15	merkezi.com	A Record	192.168.30.20

Figure 3.1.4 Mail Server



Relevant Events List:

DNS Request → DNS Reply → TCP 3-Way Handshake → HTTP GET → HTTP 200 OK → SMTP/POP3 session

Scenario 2: FTP File Transfer

Branch 2, Facility 2 – Engineer sends code files to FTP server (Branch 1, Facility 3).

Network Functionality:

The workstation establishes a TCP connection on port 21, authenticates with the FTP server, and uses the put command to transfer the file across the ISP backbone.

Protocol Data Units (PDU) Content:

Figure 3.2.1

The FTP packet is coming from a computer at branch 2, facility 2.

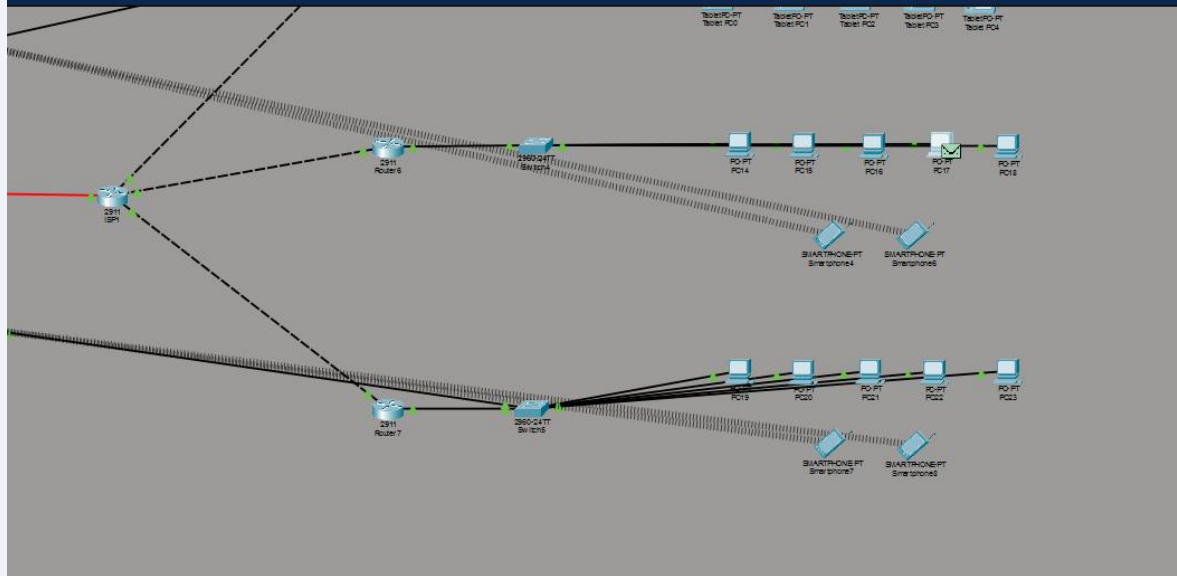


Figure 3.2.2

The packet successfully reaches one of the FTP servers at branch 1, facility 3.

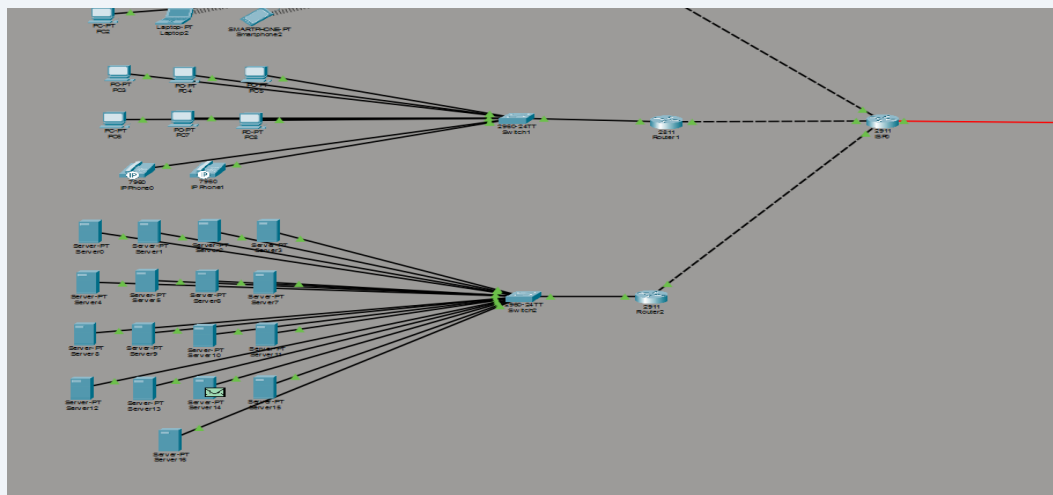
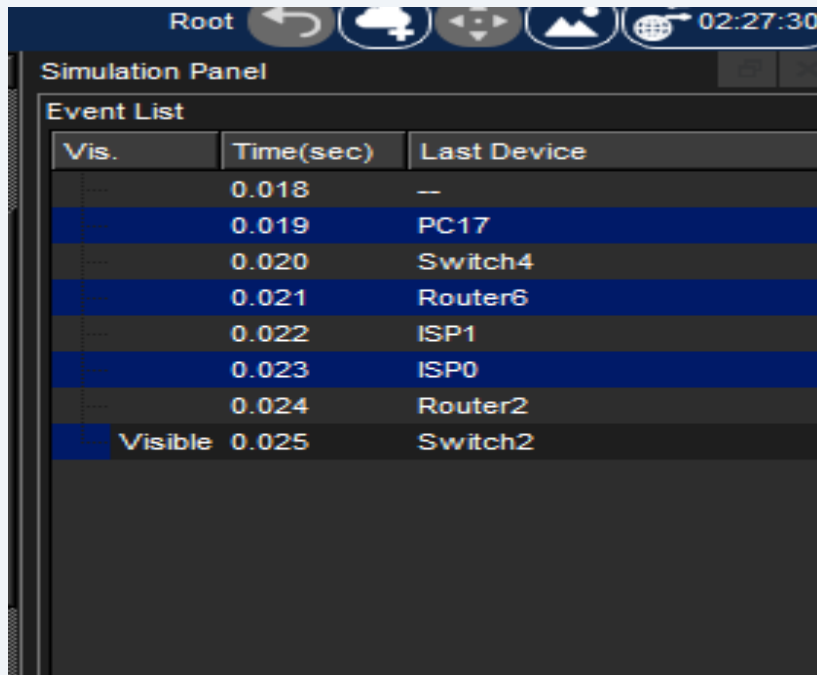


Figure 3.2.3

The routes followed by the package while reaching branch 1 facility 3.



The screenshot shows a simulation interface with a top status bar containing icons for navigation and a clock showing 02:27:30. Below this is a 'Simulation Panel' with an 'Event List' table. The table has three columns: 'Vis.', 'Time(sec)', and 'Last Device'. The rows show a sequence of events from 0.018 to 0.025 seconds, involving various network devices. The row for 0.025 seconds is highlighted in blue and includes the word 'Visible' in the 'Vis.' column.

Vis.	Time(sec)	Last Device
	0.018	—
	0.019	PC17
	0.020	Switch4
	0.021	Router6
	0.022	ISP1
	0.023	ISP0
	0.024	Router2
Visible	0.025	Switch2

Relevant Events List:

TCP SYN → FTP USER/PASS → FTP PASV → TCP Data Transfer (file chunks) → FTP 226 Transfer Complete

Scenario 3: VoIP Conference Call

Branch 1, Facility 2 – Two IP phones conduct a VoIP call.

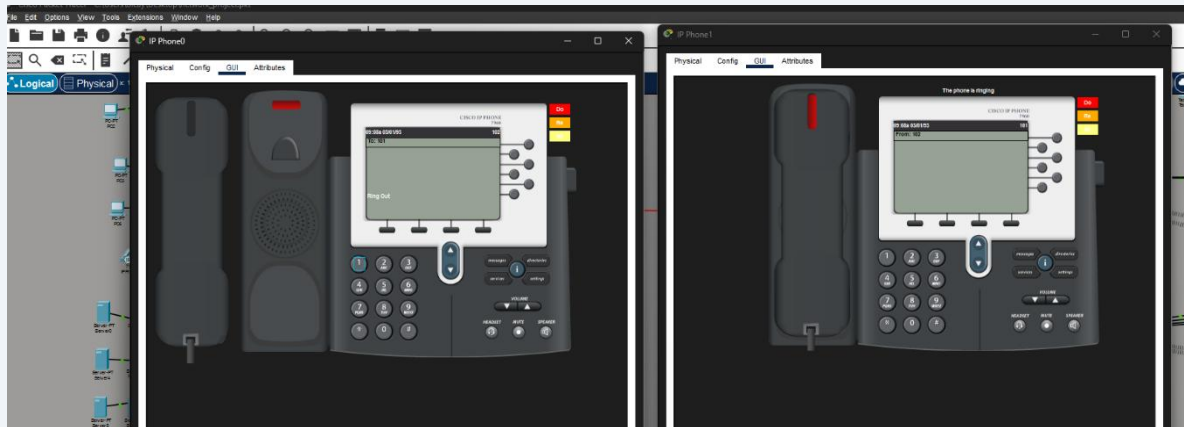
Network Functionality:

IP phones register with the Cisco Call Manager Express (CME) on the local router and establish an RTP audio stream between the two endpoints.

Protocol Data Units (PDU) Content:

Figure 3.3.1

VoIP phones successfully establish communication.



Relevant Events List:

SIP REGISTER → SIP 200 OK → SIP INVITE → SIP 200 OK → RTP stream established → SIP BYE

Scenario 4: Inter-Branch Email

Branch 1, Facility 2 → Branch 2, Facility 2 – Email communication.

Network Functionality:

The sender uses SMTP to deliver the email to the centralized Mail Server in Branch 1, Facility 3. The recipient in Branch 2 retrieves it using POP3 through the ISP backbone.

Protocol Data Units (PDU) Content:

Figure 3.4.1 Mail Server

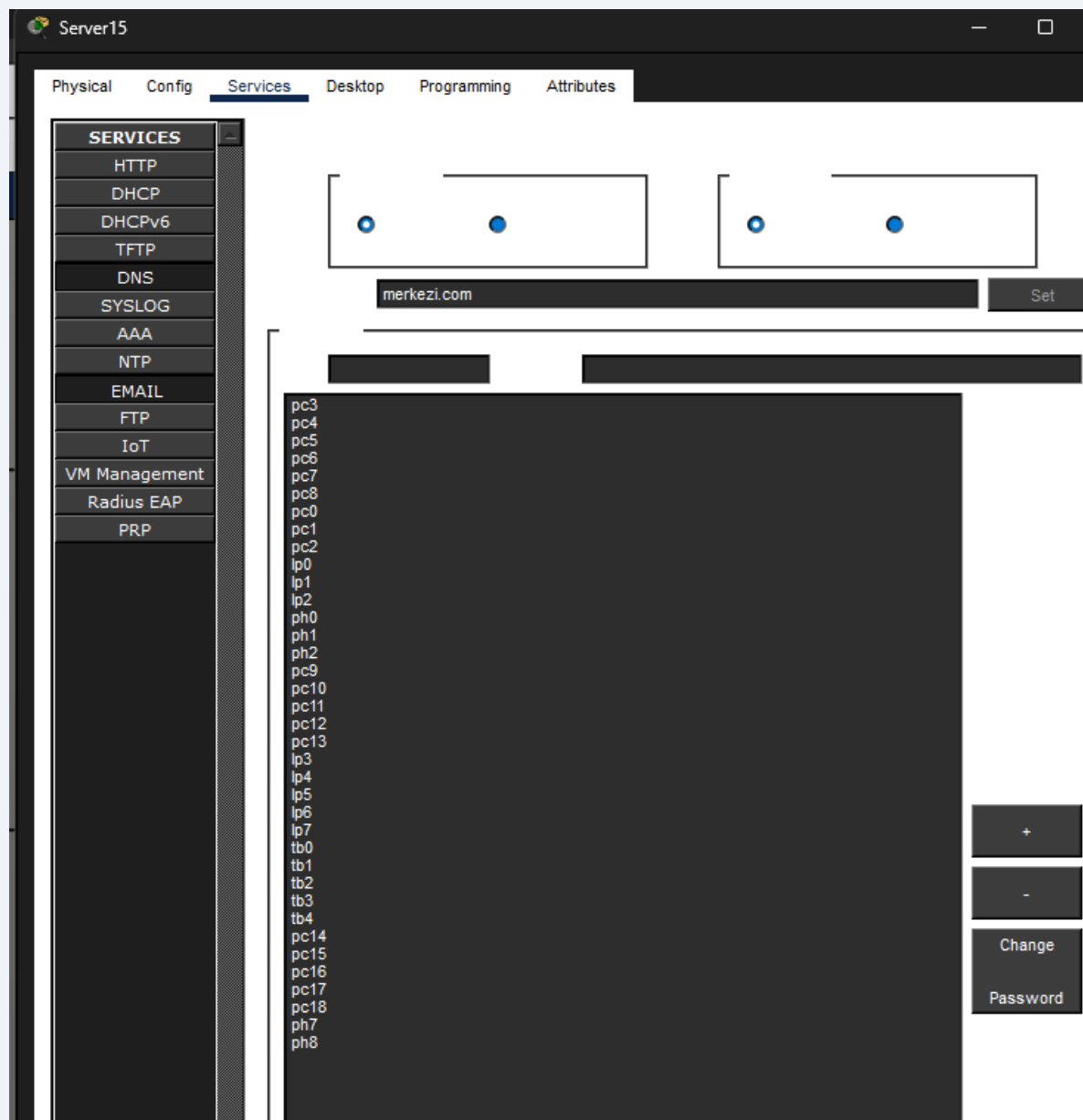


Figure 3.4.2

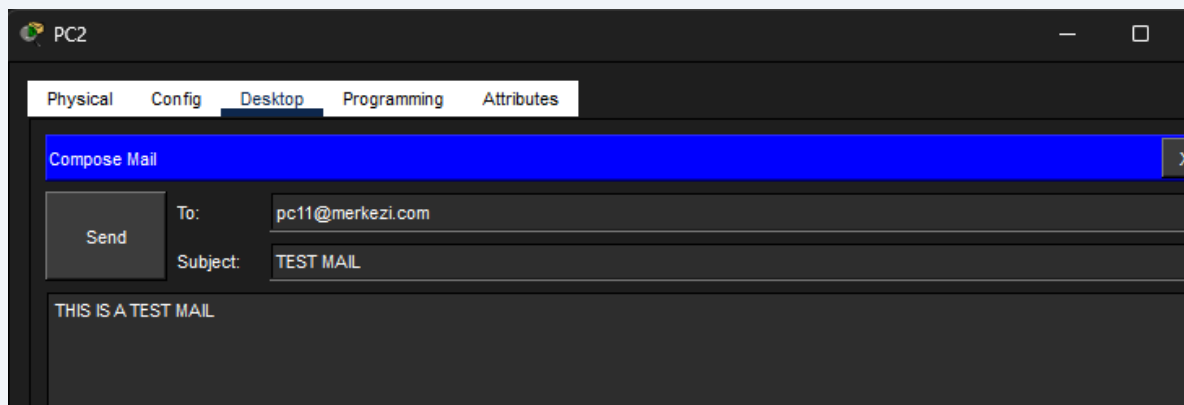


Figure 3.4.3

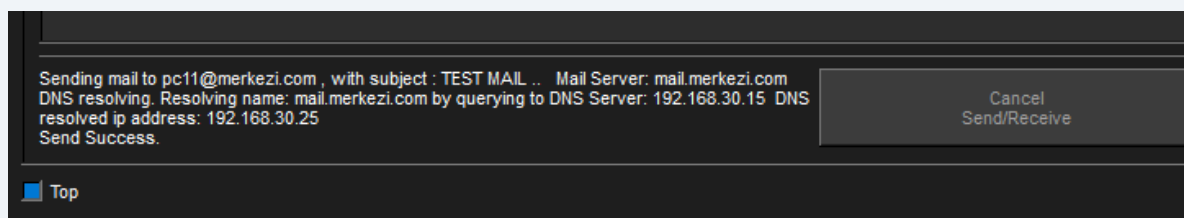


Figure 3.4.4

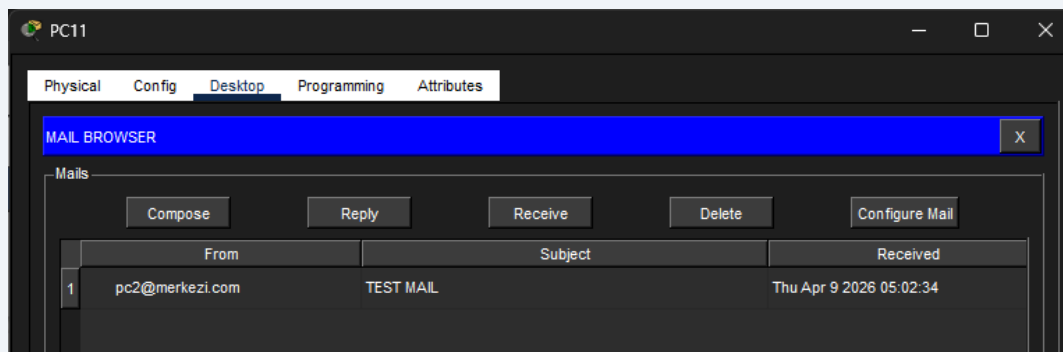
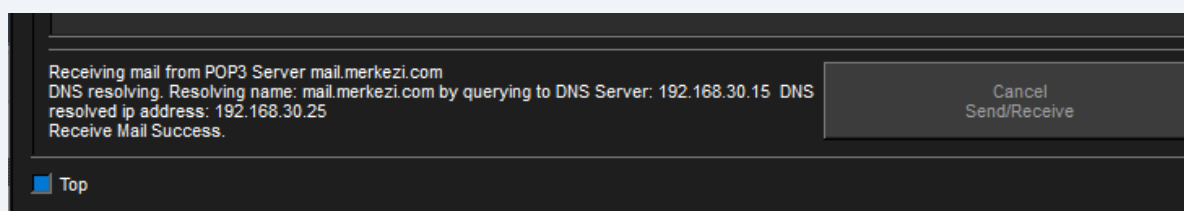


Figure 3.4.5

**Relevant Events List:**

SMTP HELO → MAIL FROM → RCPT TO → DATA → SMTP 250 OK → (Recipient) POP3
RETR → POP3 +OK

Scenario 5: ICMP Ping – Cross-Branch

Branch 2, Facility 1 pings a host in Branch 1 Facility 2.

Network Functionality:

An ICMP Echo Request is generated, routed through Branch 2 routers, traverses the ISP, and reaches Branch 1 routers, returning an ICMP Echo Reply. This validates end-to-end reachability of the MAN.

Protocol Data Units (PDU) Content:

Figure 3.5.1 Web server with IP address 192.168.30.10

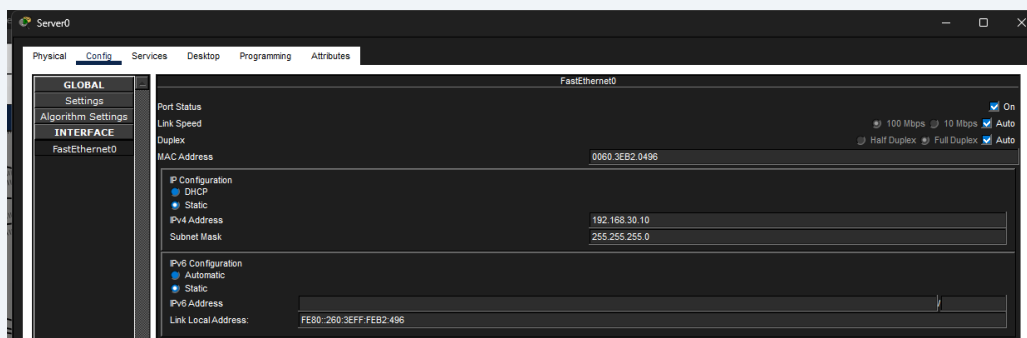


Figure 3.5.2 ping 192.168.30.10

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:
```

Figure 3.5.3 just before the packet leaves the device

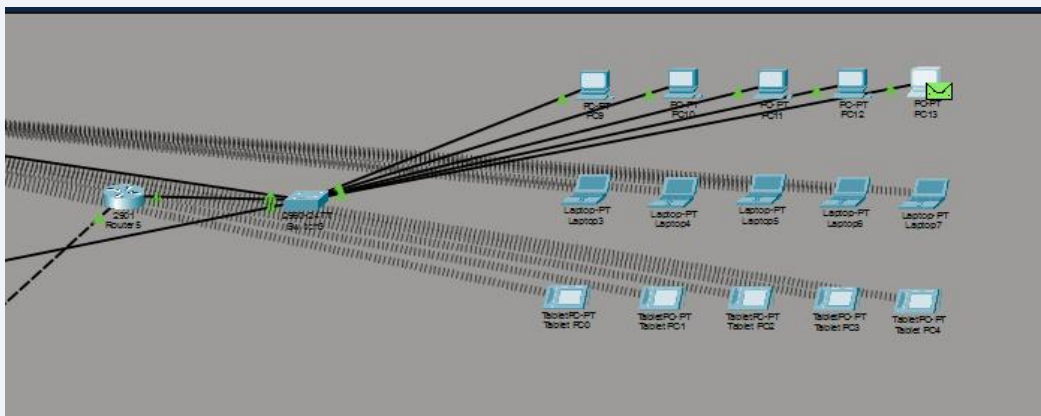
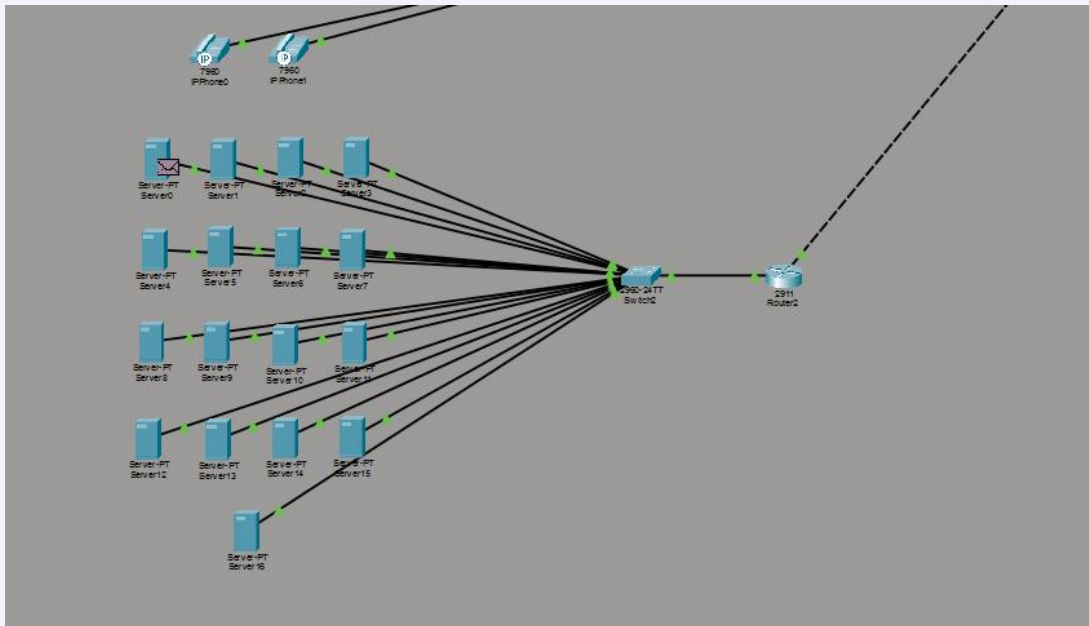


Figure 3.5.4 The packet successfully reached the web server.



Relevant Events List:

ICMP Echo Request (B2→ISP) → Router forwarding → ICMP Echo Request (ISP→B1) → ICMP Echo Reply (B1→ISP→B2)

Scenario 6: Wireless Laptop Email

Branch 1, Facility 1 – Laptop sends email to Branch 2, Facility 1.

Network Functionality:

A wireless laptop authenticates to the access point, obtains DHCP, resolves the mail server via DNS, and sends an email via SMTP across the MAN backbone. The recipient retrieves using POP3.

Protocol Data Units (PDU) Content:

Figure 3.6.1

As shown on **Figure 3.6.1** laptop3, the financial server is accessed via DNS with a domain.

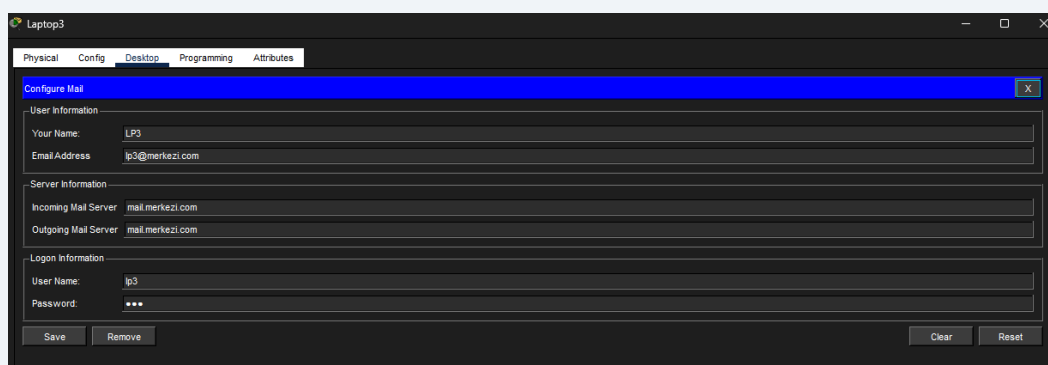


Figure 3.6.2

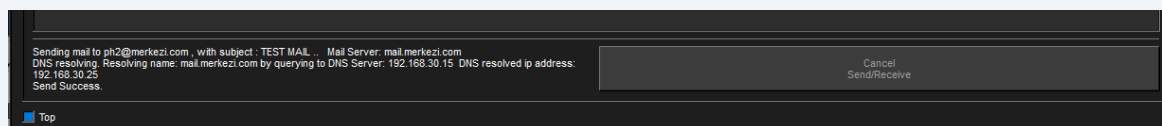
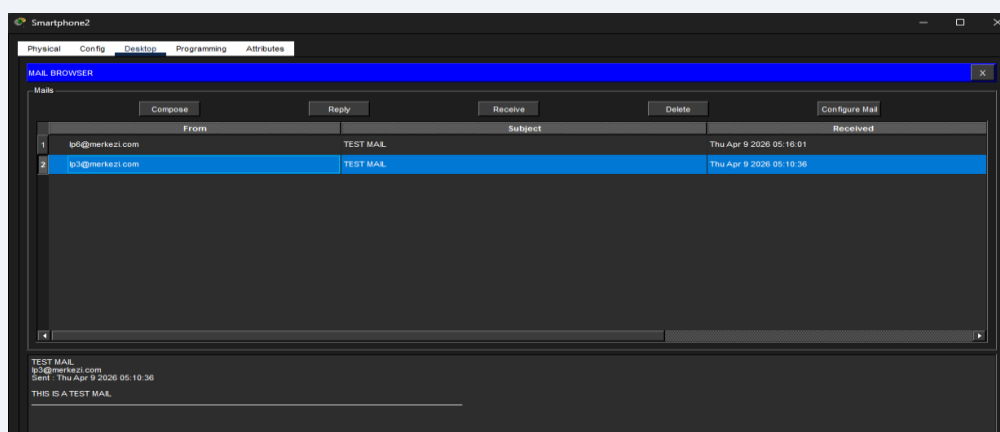


Figure 3.6.3



Relevant Events List:

802.11 Association → DHCP → DNS Resolve → SMTP → Email delivered → POP3 Retrieval

Scenario 7: SSH Remote Management

Branch 2, Facility 3 Smartphone → Web Server gateway (Branch 1, Facility 3).

Network Functionality:

The mobile device initiates a secure terminal session on port 22. SSH key exchange encrypts all traffic traversing the MAN, providing secure remote management.

Protocol Data Units (PDU) Content:

Figure 3.7.1

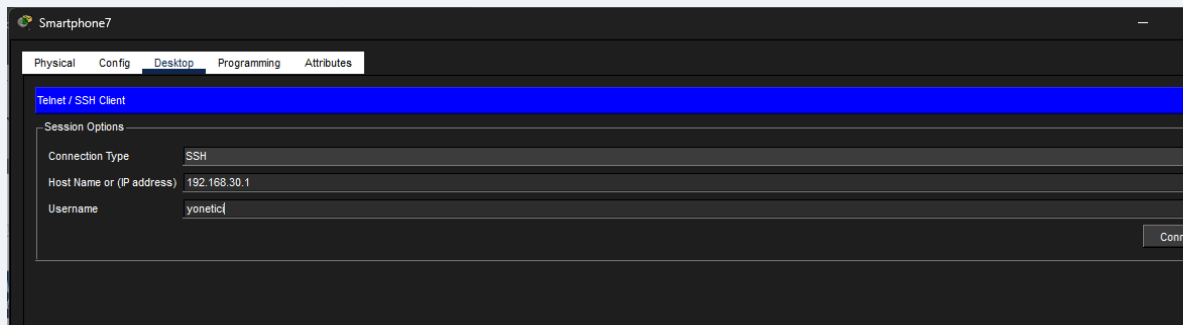
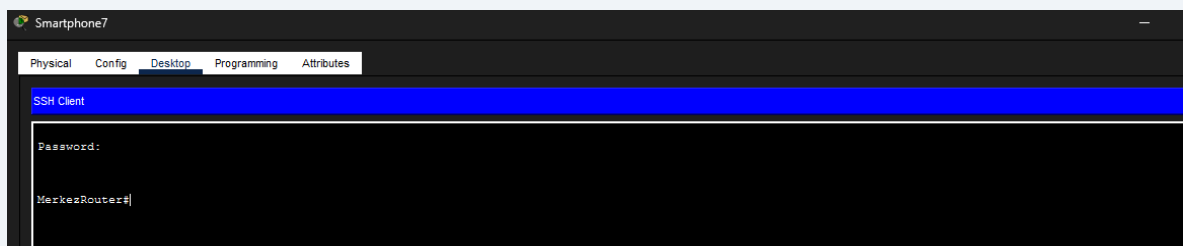


Figure 3.7.2



Relevant Events List:

TCP SYN (port 22) → TCP SYN-ACK → SSH Client Hello → SSH Key Exchange → SSH Encrypted Session

Custom Scenarios

Scenario 8: DNS Domain Resolution & Web Service Access – Branch 2, Facility 3
Custom: Validates cross-branch name resolution and web connectivity across the ISP backbone.

Network Functionality:

A workstation in the third facility of the second branch initiates a web request to [www.merkezi.com] (https://www.merkezi.com). The workstation first sends a DNS query to the centralized DNS server located in the first branch's server farm (Facility 3) to resolve the IP address. Once the IP is obtained, the workstation establishes a TCP connection with the Web server across the ISP backbone and retrieves the website data via HTTP.

Protocol Data Units (PDU) Content:

Figure 3.8.1

The device wants to access http://merkezi.com

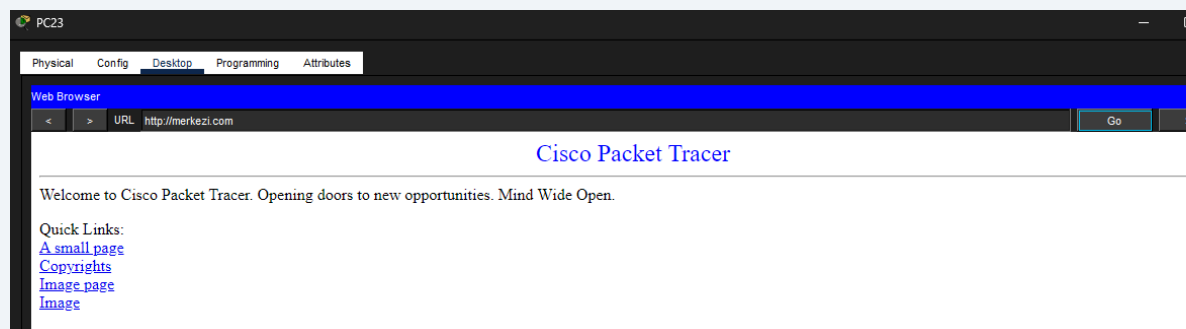


Figure 3.8.2

The packet is sent to the DNS server to learn its IP address.

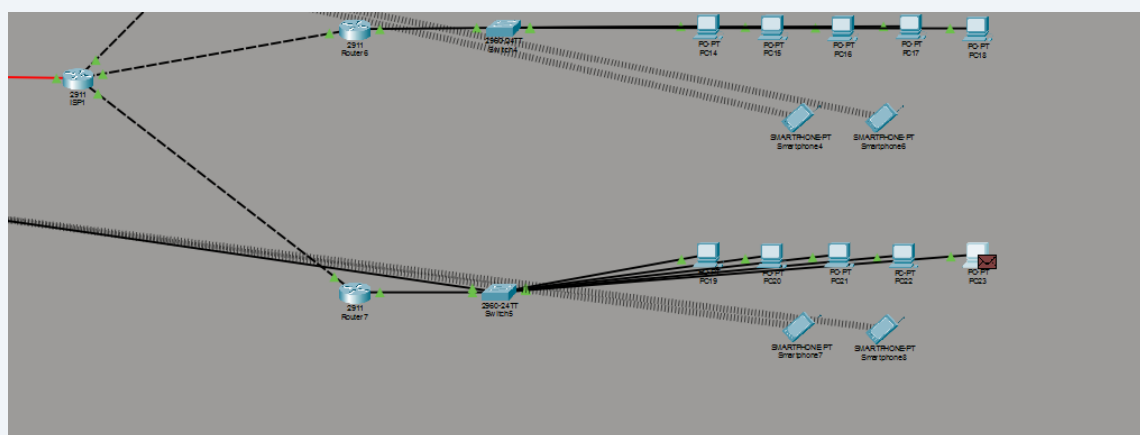
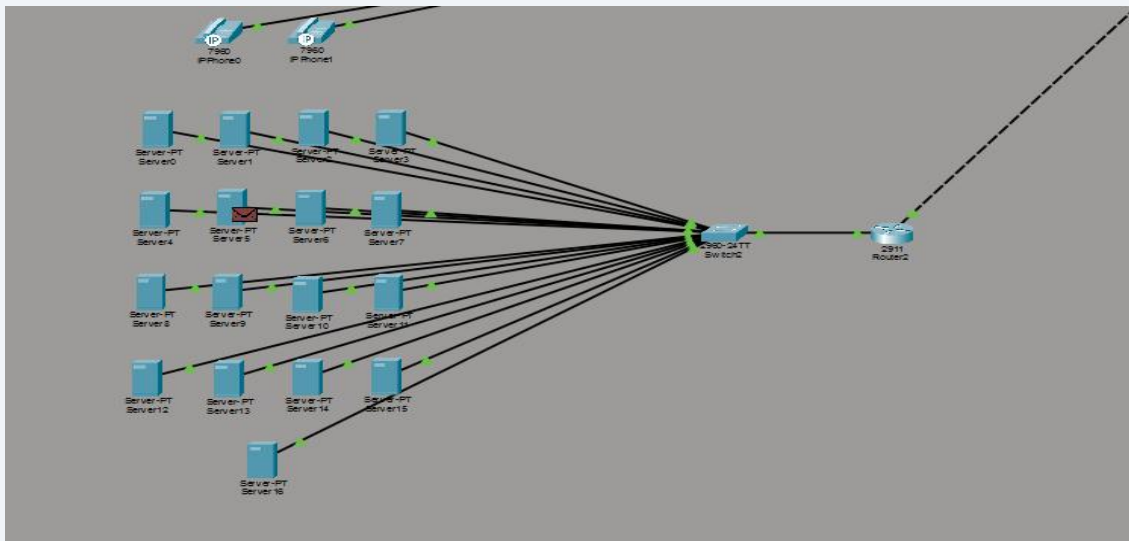
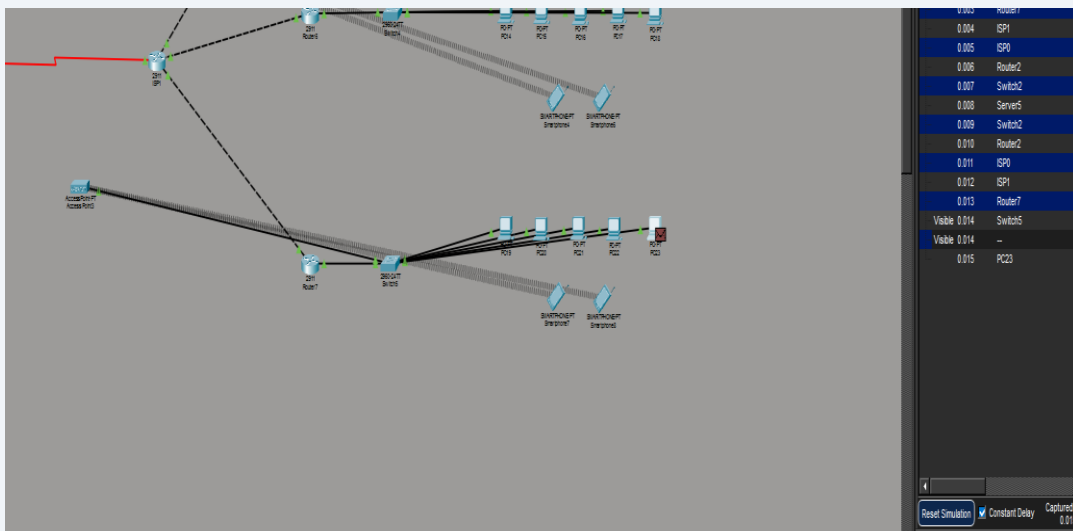


Figure 3.8.3

After the packet reaches the DNS server, the DNS server returns a non-authoritative IP address to the device.

**Figure 3.8.4**

The packet from the DNS server reaches the device, and the device learns its IP address

**Figure 3.8.4**

The packet is sent back to the web server.

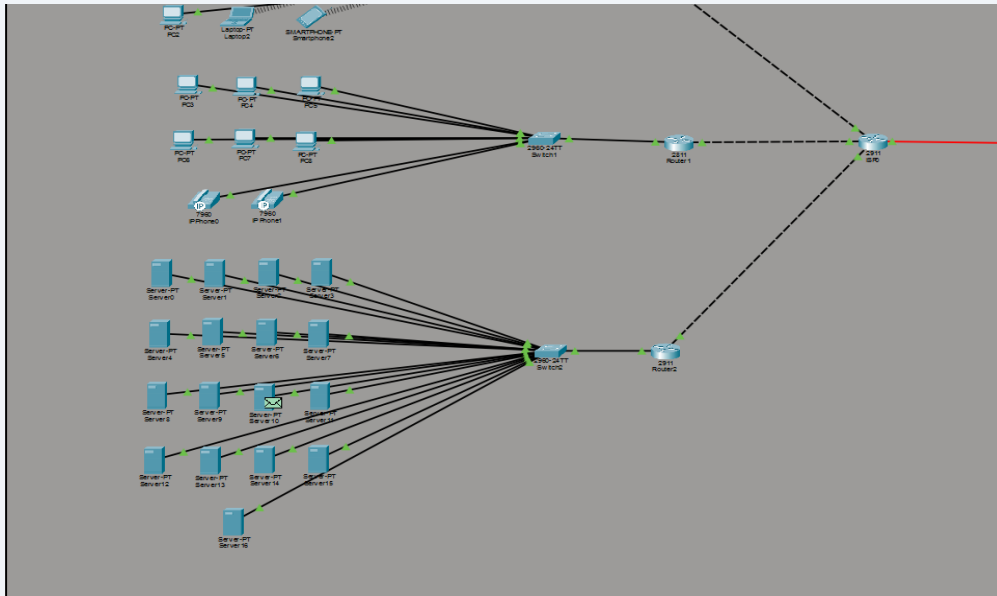


Figure 3.8.5

Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
---	0.000	---
---	0.001	PC23
---	0.002	Switch5
---	0.003	Router7
---	0.004	ISP1
---	0.005	ISP0
---	0.006	Router2
---	0.007	Switch2
---	0.008	Server5
---	0.009	Switch2
---	0.010	Router2
---	0.011	ISP0
---	0.012	ISP1
---	0.013	Router7
---	0.014	Switch5
---	0.014	---
---	0.015	PC23
---	0.016	Switch5
---	0.017	Router7
---	0.018	ISP1
---	0.019	ISP0
---	0.020	Router2
Visible	0.021	Switch2

Relevant Events List:

DNS Query (UDP 53) → DNS Answer (IP Received) → TCP SYN → TCP SYN/ACK → TCP ACK → HTTP GET → HTTP 200 OK (Success)

Scenario 9: Round-Robin DNS Load Balancing – FTP Servers

Custom: Tests DNS-based load distribution across 4 FTP servers.

Network Functionality:

A user requests ftp.domain.com. The DNS server uses round-robin records to resolve the hostname to one of the four available FTP servers in the Server Farm (Branch 1, Facility 3), demonstrating traffic load balancing without additional hardware.

Protocol Data Units (PDU) Content:

Figure 3.9.1

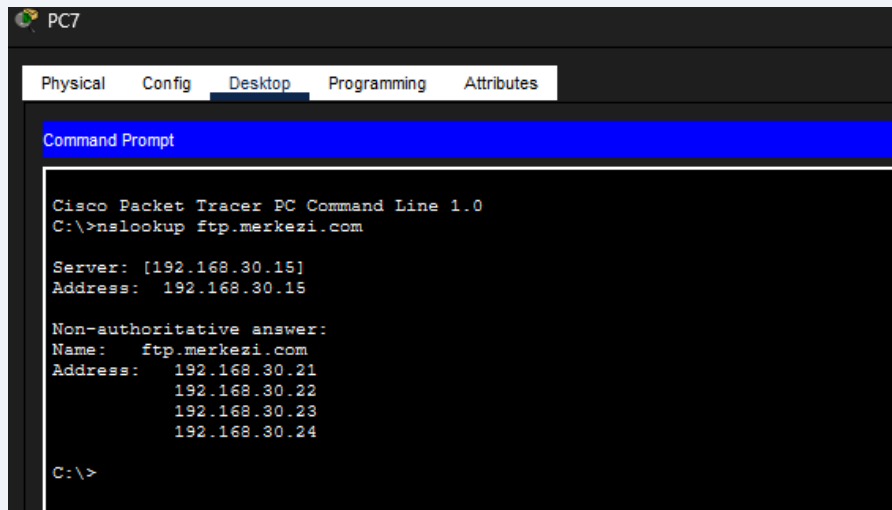


Figure 3.9.2

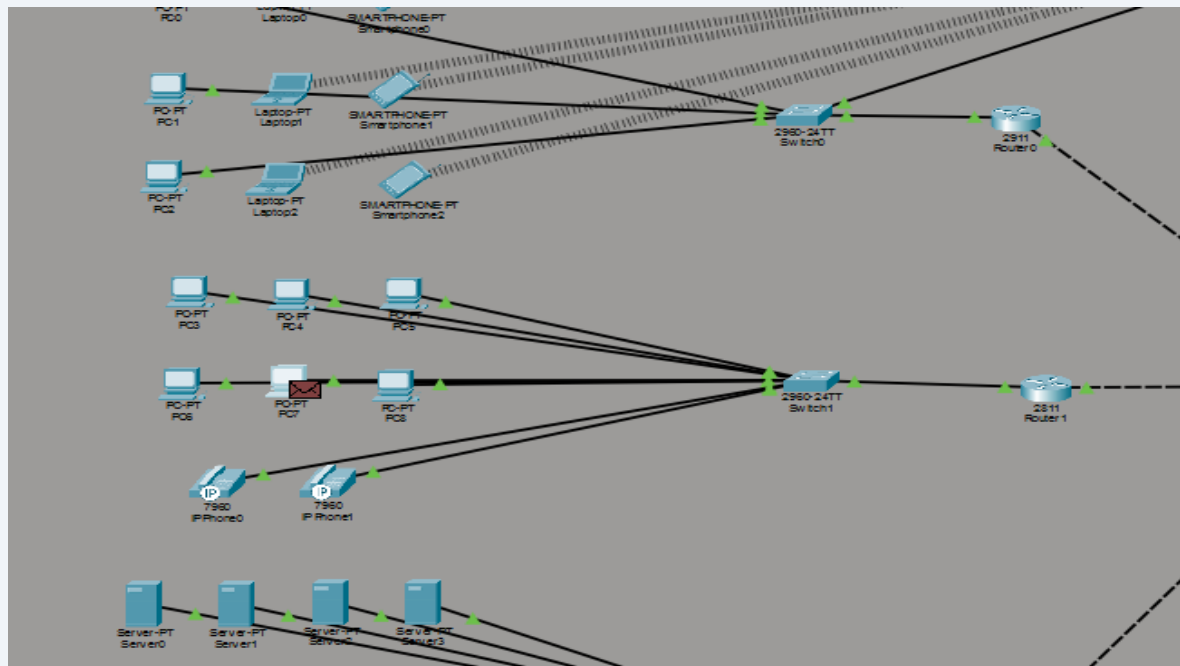


Figure 3.9.3

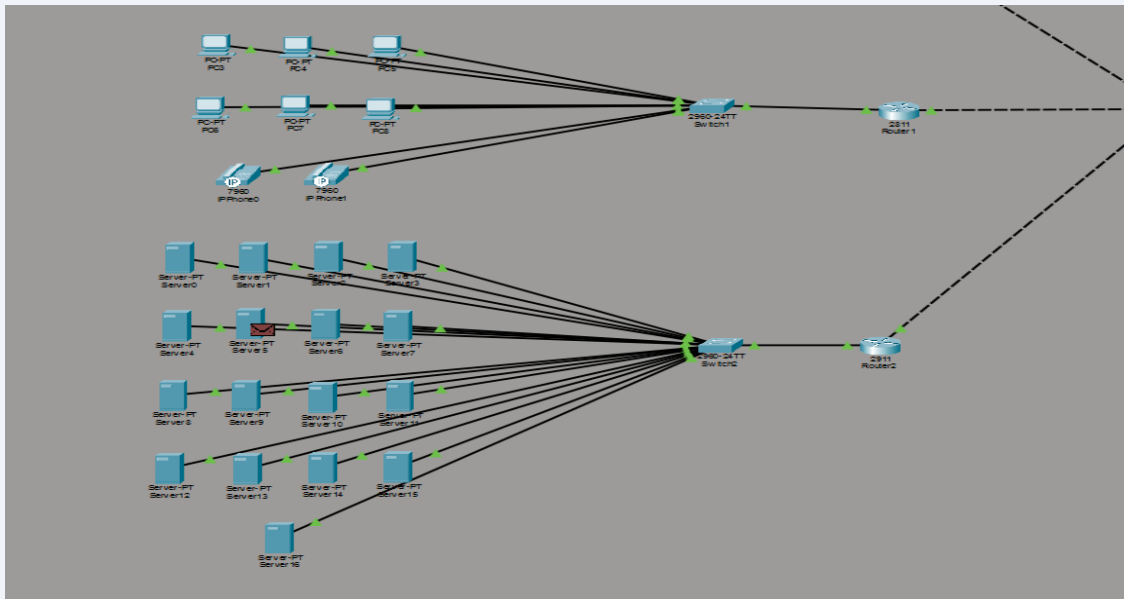
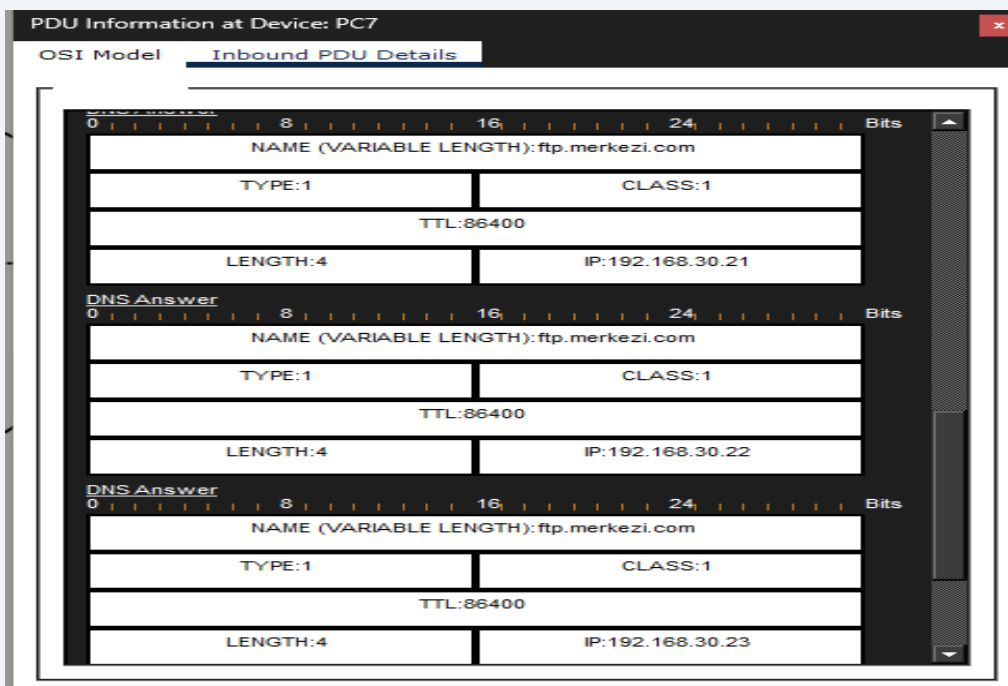


Figure 3.9.4

**Relevant Events List:**

PC generates DNS Query (ftp.domain.com) → DNS Server processes RR records → DNS Answer (IP of FTP-1/2/3/4) → FTP session established

4. Conclusion

In conclusion, the Metropolitan Area Network simulation was successfully designed and verified using Cisco Packet Tracer. The architecture fulfills all initial constraints, seamlessly bridging two distinct branch offices and their respective facilities over an ISP. By strategically implementing VLANs, systematic IP addressing, dynamic DHCP pools, and robust routing protocols, the network supports a diverse array of devices and services—from VoIP conferences to SSH remote management and FTP file transfers.

The traffic analysis of the nine scenarios confirmed that the network architecture accommodates the required traffic load with minimal delay. Future enhancements could include the implementation of advanced security protocols such as IPsec VPNs between branches over the ISP, and Quality of Service (QoS) optimizations to further prioritize VoIP traffic over best-effort data flows.

5. References

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